

AI Content Generation Project - Folder Structure

ai-content-generator/

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|
├── README.md
├── requirements.txt
├── setup.py
├── .env.example
├── .gitignore
├── docker-compose.yml
├── Dockerfile
|
├── config/
│   ├── __init__.py
│   ├── settings.py           # Main configuration
│   ├── database.py          # Database configuration
│   ├── logging.py           # Logging configuration
│   └── environments/
│       ├── __init__.py
│       ├── development.py
│       ├── production.py
│       └── testing.py
|
├── core/
│   ├── __init__.py
│   ├── exceptions.py         # Custom exceptions
│   ├── models.py             # Data models and schemas
│   ├── enums.py              # Enumerations
│   ├── constants.py          # Application constants
│   └── base/
│       ├── __init__.py
│       ├── service.py        # Base service class
│       ├── repository.py     # Base repository class
│       └── client.py         # Base client class
|
├── clients/
│   ├── __init__.py
│   ├── ai_client.py          # Google GenAI client wrapper
│   ├── database_client.py    # Database client
│   ├── cache_client.py       # Redis/Cache client
│   └── external/
│       ├── __init__.py
│       ├── translation_api.py # Translation services
│       └── storage_api.py     # File storage services
|
├── services/
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|   ├── __init__.py
|   ├── content_service.py      # Main content generation service
|   ├── prompt_service.py      # Prompt building and management
|   ├── validation_service.py   # Input validation
|   ├── cache_service.py       # Caching logic
|   ├── queue_service.py       # Job queue management
|   └── processors/
|       ├── __init__.py
|       ├── text_processor.py   # Text preprocessing
|       ├── question_processor.py
|       ├── summary_processor.py
|       ├── quiz_processor.py
|       └── explanation_processor.py
|
|── repositories/
|   ├── __init__.py
|   ├── content_repository.py   # Content CRUD operations
|   ├── user_repository.py     # User data operations
|   ├── job_repository.py      # Job/task tracking
|   └── analytics_repository.py # Usage analytics
|
|── workers/
|   ├── __init__.py
|   ├── content_worker.py      # Background content generation
|   ├── batch_worker.py        # Batch processing
|   └── scheduler.py           # Task scheduling
|
|── api/
|   ├── __init__.py
|   ├── main.py                # FastAPI application
|   ├── dependencies.py        # API dependencies
|   ├── middleware.py          # Custom middleware
|   └── routes/
|       ├── __init__.py
|       ├── content.py         # Content generation endpoints
|       ├── health.py          # Health check endpoints
|       ├── admin.py           # Admin endpoints
|       └── webhooks.py        # Webhook endpoints
|
|── utils/
|   ├── __init__.py
|   ├── helpers.py             # General helper functions
|   ├── validators.py          # Validation utilities
|   ├── formatters.py          # Text formatting utilities
|   └── rate_limiter.py        # Rate limiting utilities
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├── thread_pool.py          # Thread pool management
|
├── schemas/
|   ├── __init__.py
|   ├── request_schemas.py    # API request schemas
|   ├── response_schemas.py  # API response schemas
|   └── database_schemas.py   # Database schemas
|
├── migrations/
|   ├── __init__.py
|   └── versions/              # Database migration files
|
├── prompts/
|   ├── __init__.py
|   ├── arabic/
|   |   ├── __init__.py
|   |   ├── questions.py      # Question generation prompts
|   |   ├── summary.py        # Summary generation prompts
|   |   ├── quiz.py           # Quiz generation prompts
|   |   └── explanation.py     # Explanation prompts
|   ├── english/
|   |   ├── __init__.py
|   |   ├── questions.py
|   |   ├── summary.py
|   |   ├── quiz.py
|   |   └── explanation.py
|   └── base/
|       ├── __init__.py
|       └── prompt_templates.py # Base prompt classes
|
├── tests/
|   ├── __init__.py
|   ├── conftest.py           # Pytest configuration
|   ├── unit/
|   |   ├── __init__.py
|   |   ├── test_services/
|   |   ├── test_clients/
|   |   ├── test_repositories/
|   |   └── test_utils/
|   ├── integration/
|   |   ├── __init__.py
|   |   ├── test_api/
|   |   └── test_workflows/
|   └── fixtures/
|       ├── __init__.py

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|       |─ sample_texts.py
|       |─ mock_responses.py
|
|─ scripts/
|   |─ __init__.py
|   |─ setup_db.py           # Database setup script
|   |─ migrate.py           # Migration runner
|   |─ seed_data.py         # Sample data seeding
|   |─ deploy.py            # Deployment script
|
|─ monitoring/
|   |─ __init__.py
|   |─ metrics.py           # Application metrics
|   |─ health_checks.py     # Health monitoring
|   |─ alerts.py            # Alert configuration
|
|─ frontend/
|   |─ static/
|   |   |─ css/
|   |   |   |─ main.css
|   |   |   |─ components/
|   |   |─ js/
|   |   |   |─ main.js
|   |   |   |─ components/
|   |   |   |─ utils/
|   |   |─ images/
|   |   |   |─ icons/
|   |   |   |─ logos/
|   |   |─ fonts/
|   |─ templates/
|   |   |─ base.html
|   |   |─ index.html
|   |   |─ dashboard.html
|   |   |─ components/
|   |   |   |─ header.html
|   |   |   |─ footer.html
|   |   |   |─ forms/
|   |─ src/                 # For modern frontend frameworks
|   |   |─ components/
|   |   |─ pages/
|   |   |─ hooks/
|   |   |─ services/
|
|─ logs/                     # Log files (gitignored)
|   |─ app.log
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├─ error.log
└─ access.log

Key Folder Explanations:

Core Architecture

- `core/`: Foundation classes, models, and base abstractions
- `clients/`: External service wrappers (AI APIs, databases, cache)
- `services/`: Business logic layer with specialized processors
- `repositories/`: Data access layer

Multi-threading & Scaling

- `workers/`: Background job processors and threading logic
- `utils/thread_pool.py`: Thread pool management
- `services/queue_service.py`: Job queue for async processing

API & Integration

- `api/`: FastAPI web service layer
- `schemas/`: Request/response validation schemas
- `clients/external/`: Third-party API integrations

Content Management

- `templates/prompts/`: Organized prompt templates by language/type
- `services/processors/`: Specialized content type processors

DevOps & Monitoring

- `monitoring/`: Health checks, metrics, alerts
- `scripts/`: Automation and deployment scripts
- `migrations/`: Database schema versioning

Testing & Documentation

- `tests/`: Comprehensive test suite (unit + integration)
- `docs/`: API documentation and development guides

This structure supports:

-  **Scalability:** Clear separation of concerns
-  **Multi-threading:** Dedicated workers and thread management
-  **External integrations:** Organized client wrappers
-  **Multiple content types:** Specialized processors
-  **Testing:** Comprehensive test organization
-  **DevOps:** Docker, monitoring, and deployment ready